

CHEM 403: GENERAL CHEMISTRY LABORATORY
Summer 2018

Instructor: Dr. Samantha Reynolds

Email: Samantha.Reynolds@unh.edu

Lab Meetings: 1:00-4:00pm on Mondays, Wednesdays, and Thursdays

Laboratory: 88 Commercial St., Room 574

Course Description

The General Chemistry lab is a supplement to the General Chemistry lecture. As such, it counts as 20% of your total grade in the course. Please note that you must pass the lab in order to pass the course. There are no lab make-ups, except in the event of a school cancellation. Two or more unexcused absences from lab will result in a failing grade for the ENTIRE CHEM 403 COURSE.

You are expected to be in lab on time and prepared to start that day's experiment. This means that you have read the background and procedure for the experiment and have completed the experiment's pre-lab questions. This also means that you will enter the lab wearing appropriate clothing and shoes.

Laboratory Materials

Handouts of the procedures and pre-lab questions will be provided in lab. Copies of these handouts will also be available to view on myCourses. You will need to record and submit your data and observations for each experiment, but you will not need a dedicated laboratory notebook. A scientific calculator is also highly recommended for ease of calculations during the lab.

Appropriate clothing must be worn in the laboratory. Lab coats and goggles must also be worn while performing all experiments (unless otherwise instructed). These items can be purchased online through the UNHM bookstore, or at Work'n'Gear on South Willow St. in Manchester. "Community" goggles are available, but it is recommended that you have your own pair. Shoes must have closed toes and cover the top of the foot; no sandals or flip-flops are allowed. Suitable clothing should be worn underneath lab coats; shorts, capris and short skirts are inappropriate for lab work. If you are not dressed appropriately for lab, you will not be allowed to perform the experiment, and you will receive a zero for that experiment.

Grading

Labs will be graded on a 20-point scale:

6 points – pre-lab questions

6 points – in-lab observations

8 points – "Data and Calculations" pages and post-lab questions

Pre-lab questions must be attempted and completed before each lab; incomplete assignments at the beginning of class will receive a 10% grade deduction (2/20 points). Corrections can be made to the assignment after finishing the experiment, but students are expected to attempt this assignment prior to coming to lab. In-lab observations and the "Data and Calculations" pages along with any post-lab questions are to be completed immediately after the experiment and submitted before leaving the lab.

CHEM 403 LAB SCHEDULE
Summer 2018

Monday, 5/21	Introduction to the Chemistry Lab
Wednesday, 5/23	Exp #1. Resolution of Matter I: Paper Chromatography
Thursday, 5/24	Exp #2. The Atomic Spectrum of Hydrogen
Monday, 5/28	*No Lab* *Memorial Day*
Wednesday, 5/30	Exp #3. Resolution of Matter II: Fractional Crystallization
Thursday, 5/31	*No Lab*
Monday, 6/4	Exp #4. The Geometrical Structure of Molecules
Wednesday, 6/6	*No Lab* *Lecture Midterm*
Thursday, 6/7	Exp #5. Determination of an Unknown Chemical Formula
Monday, 6/11	Exp #6. Identification of a Compound using Mass Relationships
Wednesday, 6/13	Exp #7. Analysis of an Unknown Chloride
Thursday, 6/14	Exp #8. Spectroscopy – Using Light as a Probe
Monday, 6/18	Exp #9. Heat Effects and Calorimetry
Wednesday, 6/20	Exp #10. Molar Mass of a Gas
Thursday, 6/21	*No Lab* *Lecture Final Exam*